

Lattice Before Ring: A Two-Level Silicon Photonic Demultiplexer, Benchmarked and Tested for Fabrication Tolerance

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edX public username: wzwqoo · SiEPIC openEBL, 220 nm SOI, three cells of 605 × 410 μm

Abstract: This is the short version of the design report, written for peer review. Three layouts were submitted, each with a separate job: All three pass the EBeam DRC and SiEPIC verification with zero errors. Every device was simulated in Lumerical, and its expected spectrum is given here.

- **Chip 1 (EBeam_wzwqoo_design):** a two-level demultiplexer. An MZI lattice tree splits four coarse channels 10 nm apart, and microrings followed by contra-directional couplers (CDCs) pick four fine 200 GHz channels.
- **Chip 2 (EBeam_wzwqoo_benchmark):** the same coarse job done three other ways (an AWG, an echelle grating, a cascade of CDCs), plus a Fabry–Pérot cavity.
- **Chip 3 (EBeam_wzwqoo_FabChallenges):** copies of the chip-1 devices drawn deliberately too narrow or too wide, to measure how width errors move each filter.

Keywords: wavelength-division multiplexing, MZI lattice filter, microring resonator, contra-directional coupler, arrayed waveguide grating, fabrication tolerance

1. Summary for Reviewers

Table 1. Where each review criterion is answered

Criterion	Short answer	Section
Design concept	One system (chip 1), three alternatives to compare it with (chip 2), and width-offset copies to test its weak point (chip 3). 18 devices in total	2, 3, 4
Parameter variation	Four ring radii, two CDC periods, three cascade periods; width offsets −15 / +8 / +15 nm on rings, CDC-A and MZIs; two delay-guide widths (0.5 and 3 μm); two ring widths (500 and 479 nm)	3, 7
Calibration	GC loopback GCLoopback (chip 1); a stand-alone CDC-A (CDCA) to separate the CDC from the rings; the chip-1 ring and CDC-A are the 0 nm references for chip 3	6
Manufacturability	EBeam DRC: 0 errors on all three chips. PCell parameters were changed where the defaults would break the 70 nm Si spacing	5
Mask layout	One top cell per file, all shapes inside the floorplan; waveguides closer than 2 μm only where they are meant to couple, inside the AWG-C arm array (0.7 μm gaps) and in two 1.7 μm routing runs on chip 2	5
Design for test	Every device: input on the opt_in grating coupler, outputs on the same column at 127 μm pitch, all labels unique	6
Submission	File name = top cell name: EBeam_wzwqoo_design, EBeam_wzwqoo_benchmark, EBeam_wzwqoo_FabChallenges	5

2. Design Concept

2.1. The problem

Optical links carry many wavelengths on one fibre, and the receiver must separate them. Two kinds of filter are needed:

- a **coarse** filter that splits wide bands (here 10 nm apart) with a **flat** top, so everything inside a band passes equally;
- a **fine** filter that picks single channels 200 GHz (1.6 nm) apart with a **narrow** line.

No single filter does both well. An AWG, for example, would need at least 50 channel slots, a 370 μm star-coupler radius and about 160 arms to cover both grids, far more than the 605 μm cell holds.

2.2. The two-level architecture (chip 1)

- **Coarse level: a 1×4 MZI lattice tree (D1).** Each stage is a chain of unbalanced MZIs. Its coupler ratios shape a flat-topped passband 8.57 nm wide, wider than the 4.78 nm the fine channels occupy.
- **Fine level: four add-drop microrings (D2).** Each ring ($R \approx 7.5 \mu\text{m}$) drops one 200 GHz channel with a 0.2 nm wide line. Its FSR (12.1 nm) is larger than the 10 nm coarse spacing, so in the full system the lattice in front removes the ring's other orders.
- **Order selector on the test chip: CDCs.** On chip 1 the rings are measured without the lattice in front. A CDC after each ring passes a single band (it has no FSR), so the ring's other orders are removed on the chip as well.

Why a lattice first, and not an AWG? In simulation, an AWG coarse stage followed by fine filters loses the two outer fine channels by about 2 dB more than the inner ones. An AWG's passband is rounded (Gaussian) and only 3.0 nm wide at 1 dB, narrower than the 4.78 nm the fine channels span. The lattice's flat top covers all four.

2.3. The benchmarks (chip 2) and the width tests (chip 3)

Chip 2 does the coarse job of D1 with three other filters, on the same four channels C1–C4, so the four can be compared on one test station.

- **AWG-C (arrayed waveguide grating).** An input slab spreads the light over 27 waveguide arms, each 6.982 μm longer than the last. The growing delay tilts the wavefront leaving the arms, and the output slab focuses each wavelength onto a different output guide. It is the standard dense demultiplexer in telecom. The PDK has no AWG cell, so it was drawn from the design rules of Smit and van Dam [7].
- **EDG-C (echelle diffraction grating).** The AWG folded into one slab: instead of arms, a row of 28 etched facets reflects the light back and focuses each wavelength onto its own output [10]. A 3-pair Bragg mirror behind each facet raises its reflection [11]. It is also drawn, as the PDK has no cell.
- **CDC cascade.** Three PDK contra-directional couplers on one bus, each dropping one coarse channel and passing the rest to the next [12]. They are apodised (the grating strength tapers off at both ends) to lower the side lobes that would otherwise leak into the neighbouring channels.
- **FP cavity.** Two PDK Bragg mirrors with a half-period spacer between them pass a single narrow line inside the mirrors' stop band [6]. It asks whether a single-line filter can separate two neighbouring fine channels, which a CDC band (9 nm wide) cannot.

Chip 3 repeats the most width-sensitive devices with every drawn width, and so every gap, offset by d : -15 and $+8$ nm for the rings and CDC-A, -15 and $+15$ nm for the MZI delays. Width error is the largest fabrication variable, and on chip 1 the F2 ring line sits only 0.48 nm inside CDC-A's band.

2.4. Wavelength plan

- **Coarse channels C1–C4: 1535, 1545, 1555, 1565 nm.** The spacing is 10 nm, not the 20 nm CWDM grid, because the grating couplers lose too much at the band edges: a 20 nm grid would lose 9.96 dB on its outer channels, against 2.56 dB here.
- **Fine channels F1–F4: 1542.94, 1544.53, 1546.12, 1547.72 nm.** They are on the ITU 200 GHz grid, inside C2, the flattest part of the grating-coupler response.
- **Fine linewidth: 0.2 nm.** It gives about 10 laser points per line with the 20.4 pm test-laser step, and -24 dB between neighbouring channels.

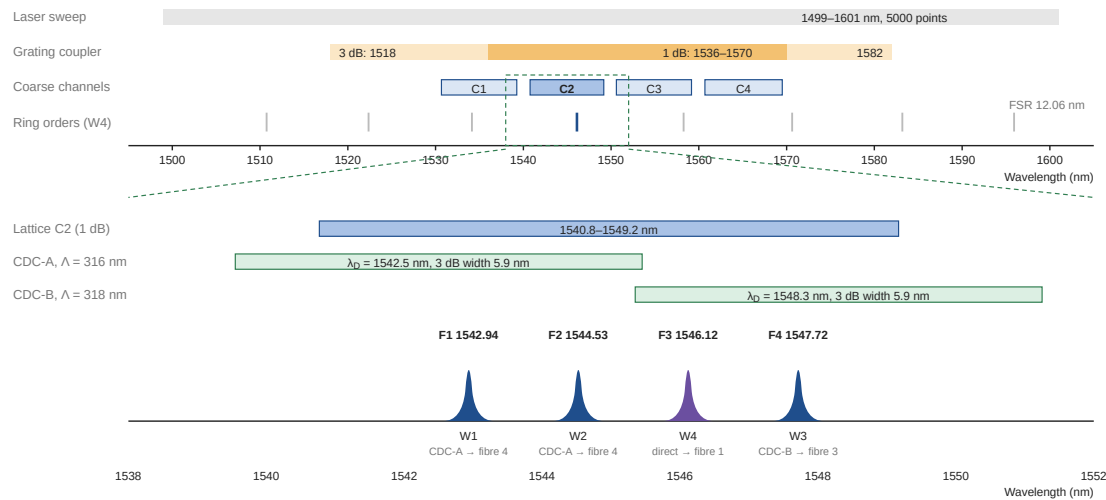


Fig. 1. Wavelength plan against the test window. Top: laser range, grating-coupler band, the coarse channels C1–C4, and the other orders of one ring. Bottom: zoom on C2 with the CDC bands and the fine channels F1–F4.

2.5. What each chip tests

Table 2. The 18 devices

Chip	Label	Device	Question it answers
1	LatticeD1	1 × 4 MZI lattice tree	Are the coarse bands flat and isolated?
1	RingCDCD2	4 rings + CDC-A + CDC-B	Do the rings land on F1–F4, and does the CDC remove their other orders?
1	CDCA	CDC-A alone	Where is the CDC band, without the rings?
1	GCLoopback	two GCs joined by a waveguide	Grating-coupler response, to normalise every spectrum
2	AWGC	1 × 4 arrayed waveguide grating	Benchmark against the lattice
2	EDGC	1 × 4 echelle grating, Bragg-mirror facets	Benchmark against the lattice
2	CDCCascade	three apodised CDCs on one bus	Benchmark against the lattice
2	FPCavity	two Bragg mirrors + half-period spacer	Can a single narrow line separate two neighbouring fine channels?
3	MZWireM15, MZWireP15	MZI, delay in 500 nm wire, width –15 / +15 nm	How far does a width error move the fringes?
3	MZWideM15, MZWideP15	the same MZI, delay in a 3 μm guide	Does a wide delay guide remove that drift?
3	CDCAM15, CDCAP8	CDC-A, width –15 / +8 nm	How far does the CDC band move?
3	RingM15, RingP8	the F2 ring (500 nm), width –15 / +8 nm	Does the ring line move with the CDC band, or away from it?
3	RingMatchM15, RingMatchP8	F2 ring redrawn at 479 nm width	Does a ring of matched width stay inside the CDC band?

The label column is the opt_in_TE_1550_device_wzwqoo_<label> text on each input grating coupler.

3. Design Equations and Parameters

Each subsection gives the formulas that size a device and the values drawn in the layout. The formulas are from Chrostowski and Hochberg [1] unless another source is given; values marked “solved” come from Lumerical (MODE FDE, EME, varFDTD or 3D FDTD).

3.1. Strip waveguide

Every device is built from the 500×220 nm strip. Its group index sets every spectral scale (FSR, linewidth), and any index change δn , from width, thickness or temperature, moves every filter by the drift law:

$$n_g = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}, \quad \delta\lambda = \frac{\lambda \delta n}{n_g} \quad (1)$$

Table 3. Strip waveguide, solved in MODE FDE

Quantity	Value
$n_{\text{eff}}(\lambda)$, λ in μm	$2.44198 - 1.12195(\lambda - 1.55) - 0.03475(\lambda - 1.55)^2$
n_g at 1550 nm	4.181 (PDK model 4.191 [8], -0.24%)
Width sensitivity	$\partial n_{\text{eff}} / \partial W = 1.574 \times 10^{-3} \text{ nm}^{-1} \rightarrow$ a filter moves 0.581 nm per nm of width, by (1)
Thickness sensitivity	$\partial n_{\text{eff}} / \partial H = 3.705 \times 10^{-3} \text{ nm}^{-1} \rightarrow$ 1.37 nm per nm of Si thickness

3.2. MZI lattice tree (D1, chip 1)

Each stage is a 2-channel half-band filter [2]: four delay sections between five directional couplers. The delay ΔL sets the stage FSR, which must be twice the channel spacing it separates (20 nm for stage 1, 40 nm for stage 2):

$$\text{FSR} = \frac{\lambda^2}{n_g \Delta L} \quad (2)$$

The integer order m then puts the passband peak exactly on the channel λ_A [9]:

$$m = \text{round} \left(\frac{n_{\text{eff}} \Delta L}{\lambda_A} \right) \Rightarrow \Delta L_{\text{drawn}} = \frac{m \lambda_A}{n_{\text{eff}}(\lambda_A)} \quad (3)$$

Each coupler of power cross-coupling K is the PDK ebeam_dc_te1550 (two 500 nm guides, gap 200 nm) with a straight length L_c :

$$K = \sin^2 \left(\frac{\pi(L_c + L_0)}{2L_x} \right) \quad (4)$$

where L_x is the cross-over length and L_0 the coupling already done in the S-bends.

Table 4. Lattice parameters

Parameter	Value
Delays by (2), (3): stage 1 / 2A / 2B	FSR 20 / 40 / 40 nm; $\lambda_A = 1535 / 1535 / 1545$ nm; $m = 46 / 23 / 23$; $\Delta L = 28.741 / 14.370 / 14.530$ μm
π phase offset in the third section	$L_\pi = \lambda_A / (2n_{\text{eff}}) = 0.312\text{--}0.316$ μm
Coupler ratios $K_0 \dots K_4$	0.500, 0.0730, 0.0658, 0.500, 0.2154
L_x , L_0 at 1550 nm (solved, 3D FDTD of the PDK cell)	38.29 μm , 2.04 μm
Straight lengths L_c , by (4)	17.106, 4.628, 4.284, 17.106, 9.724 μm

Why these coupler ratios. The published set of Dwivedi et al. [3] (0.50, 0.13, 0.12, 0.50, 0.25) leaks -14.1 dB of the other channel within ± 3 nm of each channel centre. The set above was found by minimising that worst leak and reaches -25.9 dB, with an 8.57 nm flat top.

3.3. Microrings (D2, chip 1)

Four add-drop rings on one bus, each built from two PDK half-rings (ebeam_dc_halfiring_straight, point-coupled). The radius sets the FSR and the channel; the gap sets the coupling κ^2 and so the linewidth [4]:

$$\text{FSR} = \frac{\lambda^2}{n_g 2\pi R}, \quad \text{FWHM} \approx \frac{\text{FSR} \kappa^2}{\pi} \quad (5)$$

A ring's Lorentzian line leaks into a neighbouring channel $\Delta\lambda$ away by

$$\text{leak} = \frac{1}{1 + (2\Delta\lambda/\text{FWHM})^2} = \frac{1}{1 + (2 \times 1.593/0.2)^2} = -24.1 \text{ dB} \quad (6)$$

Table 5. Ring parameters

Ring	Channel	Radius	Bent n_{eff} (solved)	FSR	Drop routed to
W1	F1, 1542.94 nm	7.5157 μm	2.45053	12.056 nm	CDC-A $\rightarrow$ fibre 4
W2	F2, 1544.53 nm	7.5289 μm	2.44874	12.059 nm	CDC-A $\rightarrow$ fibre 4
W4	F3, 1546.12 nm	7.5422 μm	2.44695	12.062 nm	fibre 1, direct
W3	F4, 1547.72 nm	7.5555 μm	2.44516	12.066 nm	CDC-B $\rightarrow$ fibre 3

All four rings: width 500 nm, order $m = 75$, gap 110 nm for $\kappa^2 = 0.049$ per coupler (FWHM 0.2 nm with 3 dB/cm loss assumed; drop loss -0.28 dB). One 200 GHz channel step needs 13.25 nm more radius. The rings are tuned by radius rather than width, because on the 1 nm layout grid a width step moves a channel by 36 % of a spacing and a radius step by only 7.5 %.

3.4. Contra-directional couplers CDC-A and CDC-B (chip 1)

The PDK contra_directional_coupler couples a forward mode in one guide to a backward mode in the other at one band, dropped back towards the input side [5]. Its centre and each guide's own (unwanted) Bragg reflection are

$$\lambda_D = (n_1 + n_2) \Lambda, \quad \lambda_{B,i} = 2 n_i \Lambda \quad (7)$$

and its peak drop grows with the grating length L as $\tanh^2(\kappa L)$.

Table 6. CDC parameters

Parameter	CDC-A	CDC-B
Guides, corrugation	560 / 440 nm wide, teeth 50 / 25 nm	same
Gap	116 nm (PDK default 100 nm puts facing teeth 62.5 nm apart, below the 70 nm minimum Si space)	same
Period Λ	316 nm	318 nm
Periods N , length, apodisation	300, 94.8 μm , none (uniform)	same
λ_D by (7), with the solved $n_1 + n_2$	1542.54 nm ($n_1 + n_2 = 4.8815$)	1548.28 nm
Self-Bragg $\lambda_{B1} / \lambda_{B2}$ by (7)	1579.72 / 1508.70 nm	1585.88 / 1514.06 nm
Grating strength κ (solved, EME)	$\approx 0.040 \mu\text{m}^{-1} \rightarrow \kappa L = 3.8$, peak drop 0.998	same grating, solved for CDC-A only
Channels carried	F1 and F2	F4

3.5. FP cavity (chip 2)

Two PDK Bragg gratings (ebeam_bragg_te1550) act as mirrors. A half-period spacer between them opens one transmission line at the centre of their stop band [6]. The line narrows as the mirror reflection approaches 1:

$$\lambda_B = 2\bar{n} \Lambda, \quad R = \tanh^2(\kappa L), \quad L_s = \frac{\Lambda}{2} \quad (8)$$

The target line width is 0.2–0.32 nm: narrow enough to reject F1 by 20 dB, no narrower than the 0.2 nm ring line. 60-period mirrors ($R = 0.46$) give no usable line, and 200 periods narrow it to about 0.1 nm. 150 periods give 0.30 nm and reject F1 by 20.5 dB.

Table 7. FP cavity parameters

Parameter	Value
Mirror grating	$\Lambda = 317$ nm, corrugation $\Delta W = 50$ nm, 500 nm guide
Grating strength κ (solved, EME of one 60-period mirror)	$0.043 \mu\text{m}^{-1}$ ($\lambda_B = 1544.3$ nm, so $\bar{n} = 2.436$ by (8))
Periods per mirror	150 ($47.55 \mu\text{m}$) $\rightarrow R = 0.94$ by (8), line width 0.30 nm
Spacer L_s	158 nm ($\Lambda/2$ on the 1 nm grid)
Layout	one straight vertical piece, $95.3 \mu\text{m}$ long: a bend would stretch the spacer and open more lines

3.6. AWG-C (chip 2)

Three rules size an AWG [7]. The arm step ΔL sets the grating order and the FSR, the slab radius R_a sets the spacing of the channels at the output, and the arm count M must capture the beam that spreads from the input guide (half-width θ_0):

$$m = \frac{n_{\text{eff}} \Delta L}{\lambda_c}, \quad \text{FSR} = \frac{\lambda_c^2}{n_g \Delta L} \quad (9)$$

$$R_a = \frac{d_o n_s d \lambda_c}{n_g \Delta L \Delta \lambda_{\text{ch}}} \quad (10)$$

$$M \geq \frac{3 \theta_0 R_a}{d} \quad (11)$$

Table 8. AWG-C parameters

Parameter	Value
Centre wavelength, channel spacing, channels	$\lambda_c = 1550$ nm, $\Delta \lambda_{\text{ch}} = 10$ nm, 4 outputs
Order m , arm step ΔL , FSR, by (9)	11, $6.982 \mu\text{m}$, 82.3 nm
Slab index n_s (solved, slab FDE)	2.8446
Arm pitch d , output pitch d_o	$2.0 \mu\text{m}$, $2.0 \mu\text{m}$
Slab radius R_a , by (10)	$60.44 \mu\text{m}$
Beam half-width θ_0 , arm count M , by (11)	0.29 rad, 27 arms
Arm and I/O tapers	$0.5 \rightarrow 1.8 \mu\text{m}$ over $10 \mu\text{m}$, so $0.2 \mu\text{m}$ gaps where they meet the slab
Refocus (solved, varFDTD)	every I/O aperture moved $1.6 \mu\text{m}$ towards the grating, because the beam leaves the taper from a point $1.6 \mu\text{m}$ behind its mouth
Arm layout	27 nested-U arms, $161\text{--}342 \mu\text{m}$ long in steps of ΔL ; $1.2 \mu\text{m}$ pitch ($0.7 \mu\text{m}$ gaps) at the U tops and legs, coupling estimated at about -35 dB
Cell size	$139 \times 197 \mu\text{m}$

3.7. EDG-C (chip 2)

The facets lie on a grating circle. The grating equation sets the facet pitch p for order m , and the circle radius R sets the channel spacing at the outputs [10]:

$$n_s p (\sin \theta_i + \sin \theta_d) = m \lambda, \quad \text{FSR} = \frac{\lambda n_s}{m n_{gs}} \quad (12)$$

$$R \approx \frac{d_o n_s \lambda_c}{2 n_{gs} \Delta \lambda_{\text{ch}} \sin \theta_B} \quad (13)$$

Table 9. EDG-C parameters

Parameter	Value
Slab index n_s , slab group index n_{gs} (solved)	2.846, 3.695
Order m , FSR, by (12)	14, 85.3 nm
Blaze angle θ_B , facet pitch p	45° , 5.39 μm
Grating-circle radius R , by (13); output pitch d_o	168.9 μm ; 2.0 μm
Facets	28
Facet mirrors	3 quarter-wave pairs behind each facet: Si 136 nm, oxide 268 nm [11]
Cell size	$108 \times 191 \mu\text{m}$

3.8. CDC cascade (chip 2)

Three PDK CDCs as in Section 3.4, sized by (7), one per channel, on one bus [12].

Table 10. CDC cascade parameters

Parameter	Value
Periods Λ (channel)	325 nm (C4), 321 nm (C3), 318 nm (C2); C1's CDC is left out, as only three output fibres exist
Periods N , length, gap	250, about 80 μm , 116 nm
Apodisation	PCell Gaussian index $a = 1.3$: every tooth at least 5 nm deep (an assumed print limit); largest side lobe -13.2 to -13.5 dB, against -4.1 to -4.7 dB uniform
Bus order	longest wavelength first, so no channel reaches a CDC whose self-Bragg reflection (about 37 nm above its drop) could send it back

3.9. Width-offset copies (chip 3)

Every width is offset by d . A wider guide also narrows the gap beside it, so the gaps change by $-d$. Table 11 lists the values drawn in the GDS.

Table 11. Chip-3 drawn values

Label	Copy of	d	Drawn widths and gap
CDCAM15 / CDCAP8	CDC-A (560 / 440 nm, gap 116 nm)	$-15 / +8$ nm	545 / 425 nm, gap 131 nm / 568 / 448 nm, gap 108 nm
RingM15 / RingP8	F2 ring W2 (500 nm, gap 110 nm, $R = 7.529 \mu\text{m}$)	$-15 / +8$ nm	485 nm, gap 125 nm / 508 nm, gap 102 nm
RingMatchM15 / RingMatchP8	F2 ring at $W^* = 479$ nm, $R = 7.639 \mu\text{m}$	$-15 / +8$ nm	464 nm, gap 125 nm / 487 nm, gap 102 nm
MZWireM15 / MZWireP15	MZI, delay in 500 nm wire	$-15 / +15$ nm	delay legs 485 / 515 nm
MZWideM15 / MZWideP15	MZI, delay in 3 μm guide	$-15 / +15$ nm	delay legs 2.985 / 3.015 μm

The MZIs. Each is built from PDK cells: an ebeam_y_1550 Y-branch as splitter and another as combiner. Each arm has four 500 nm bends ($R = 5 \mu\text{m}$) and two vertical delay legs. A leg is a 40 μm ebeam_taper_te1550 from 0.5 μm to the leg width, a straight, and a taper back. The arms are mirror images; the upper arm's straights are 7 μm and the lower arm's 2 μm , so $\Delta L = 2 \times (7 - 2) = 10 \mu\text{m}$, all of it in the leg width.

The matched ring W^* . The ring line moves 0.591 nm per nm of width, but CDC-A's band moves 0.667 nm/nm, because one CDC guide is only 440 nm wide and the gap shrinks as d grows. A 479 nm ring moves at 0.662 nm/nm, nearly with the CDC. Its radius grows from 7.529 to 7.639 μm to keep F2 at the same order.

4. Simulated Performance

Each component was solved in Lumerical (MODE FDE, EME, varFDTD or 3D FDTD), exported as S-parameters and assembled in INTERCONNECT with the PDK grating couplers. Table 12 is the expected measurement, with grating-coupler loss removed.

Table 12. Expected results per device

Label	Expected result
LatticeD1	Peak loss 0.36–0.49 dB; 1 dB band 8.4–8.9 nm; worst leak of another channel within ± 3 nm: -25.3 / -21.8 / -19.1 / -15.1 dB for C1 / C2 / C3 / C4
RingCDCD2	F1–F4 lines at 1542.93 / 1544.53 / 1546.11 / 1547.72 nm, 0.16–0.20 nm wide, peak -0.31 to -0.54 dB. Isolation about 30 dB on the CDC outputs, -24.1 dB on the direct ring drop
CDCA	1 dB band 1536.4–1545.0 nm, peak -0.05 dB at 1542.8 nm
GCLoopback	Two grating couplers: -4.71 dB peak at 1554.5 nm
AWGC	Peak -2.37 to -2.60 dB; 1 dB band 2.8–3.0 nm; worst leak within ± 3 nm -24.5 to -27.8 dB
EDGC	Peak -1.24 to -2.33 dB; isolation within ± 3 nm 19.9–32.3 dB
CDCCascade	Peak -0.09 to -0.16 dB; 1 dB band 7.0–8.1 nm; worst leak within ± 3 nm -16.5 to -20.2 dB
FPCavity	One line at 1544.28 nm, 0.30 nm wide, -0.33 dB (-5.51 dB through the two grating couplers)
Chip 3	See Table 14

Main comparisons.

- **Coarse filters.** The lattice has the widest flat top (8.4–8.9 nm at 1 dB) and low loss. The AWG and the EDG isolate better but have rounded passbands only 2.8–4.0 nm wide. The CDC cascade loses least but leaks most (-16.5 dB).
- **Fine filters.** A plain ring drop leaks -24.1 dB into the neighbouring channel, as (6) predicts for a 0.2 nm line. A CDC after the ring removes the ring's other orders, not this neighbour leak.
- **FP cavity.** Its 0.30 nm line rejects F1 by 20.5 dB, so it can separate two neighbouring fine channels that a CDC band (9 nm wide) cannot.

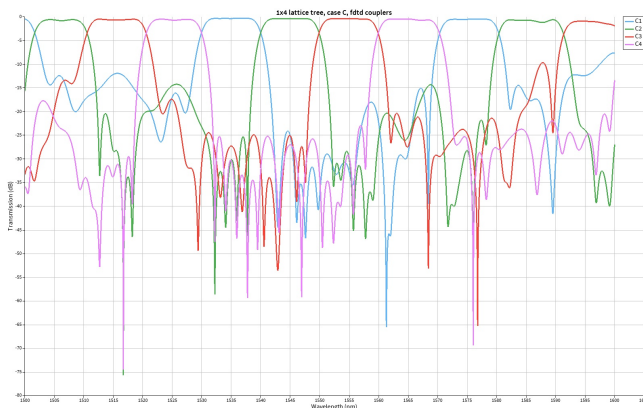


Fig. 2. Lattice tree D1, outputs C1–C4, couplers from 3D FDTD (INTERCONNECT).

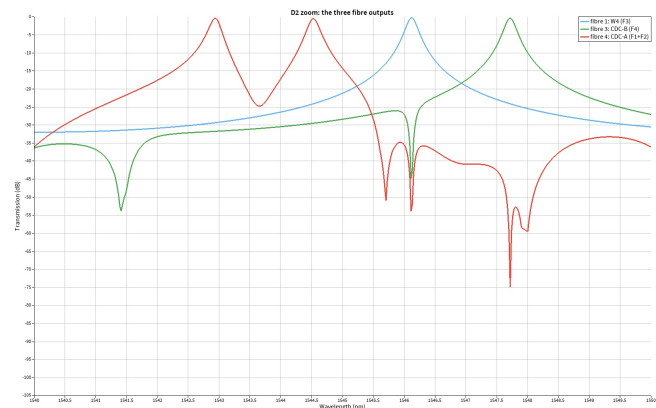


Fig. 3. Ring bank D2, 1540–1550 nm: W4 direct drop (fibre 1), CDC-B (fibre 3), CDC-A (fibre 4).

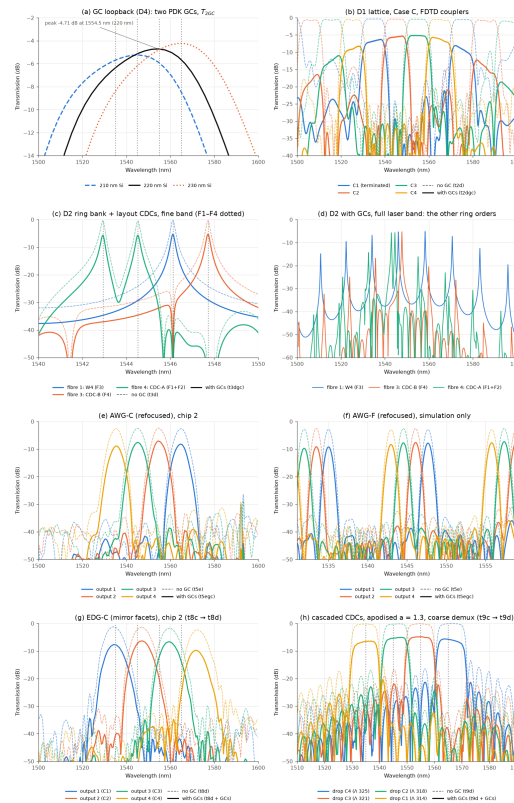


Fig. 4. What the fibre measurement will see: each device without (dashed) and with (solid) its two PDK grating couplers. (a) GC loopback at 210 / 220 / 230 nm Si. (b) Lattice D1. (c) Ring bank D2, fine band. (d) D2 over the full laser band. (e) AWG-C and (f) AWG-F (simulation only). (g) EDG-C. (h) CDC cascade.

5. Layout and Verification

5.1. Layout

Manufacturability choices. Every component is a PDK cell or PCell, with only its parameters changed:

- **CDC gap 116 nm instead of the default 100 nm.** At 100 nm the facing teeth are only 62.5 nm apart, below the 70 nm minimum Si spacing; at 116 nm they are 78.5 nm apart.
- **CDC-A and CDC-B unapodised.** The default apodisation tapers the end teeth to sub-nm depths, too shallow to print.
- **Cascade CDCs apodised at $a = 1.3$,** the strongest taper that keeps every tooth at least 5 nm deep (an assumed print limit, not a PDK rule; the case where those teeth do not print is checked below).
- **Ring gap 110 nm and coupler gap 200 nm,** both above the 70 nm minimum.
- **FP cavity drawn as one straight piece,** so the spacer keeps its 158 nm length (a bend would stretch it and open extra lines).
- **Odd widths are drawn 1 nm narrower.** The layout grid is 1 nm, and the PDK cells place each waveguide edge at $\pm$ width/2 from its centre. For an odd width both edges land on a half-nanometre (487 nm $\rightarrow \pm 243.5$ nm) and both round toward the centre, so 487 nm is drawn as **486 nm**; even widths are exact. Only chip 3's width-offset copies have odd widths; Section 5.2 lists what this does to their offsets.

Close waveguides. Silicon edges closer than 2 μ m occur only where light is meant to couple (lattice couplers, ring gaps, CDC and Bragg teeth, the AWG and EDG tapers at their slabs) and in two places on chip 2:

- **AWG-C arms:** 1.2 μ m pitch (0.7 μ m gaps) at the U tops and legs, coupling between neighbours estimated at about -35 dB over the 60 μ m U tops (not simulated).
- **Two routing pairs** run 1.7–1.8 μ m apart for about 21–24 μ m. At that gap two 500 nm strips exchange a negligible fraction of their power.

Fibre-array map. The fibre array has one input and three outputs at 127 μ m pitch. Table 13 lists what each fibre

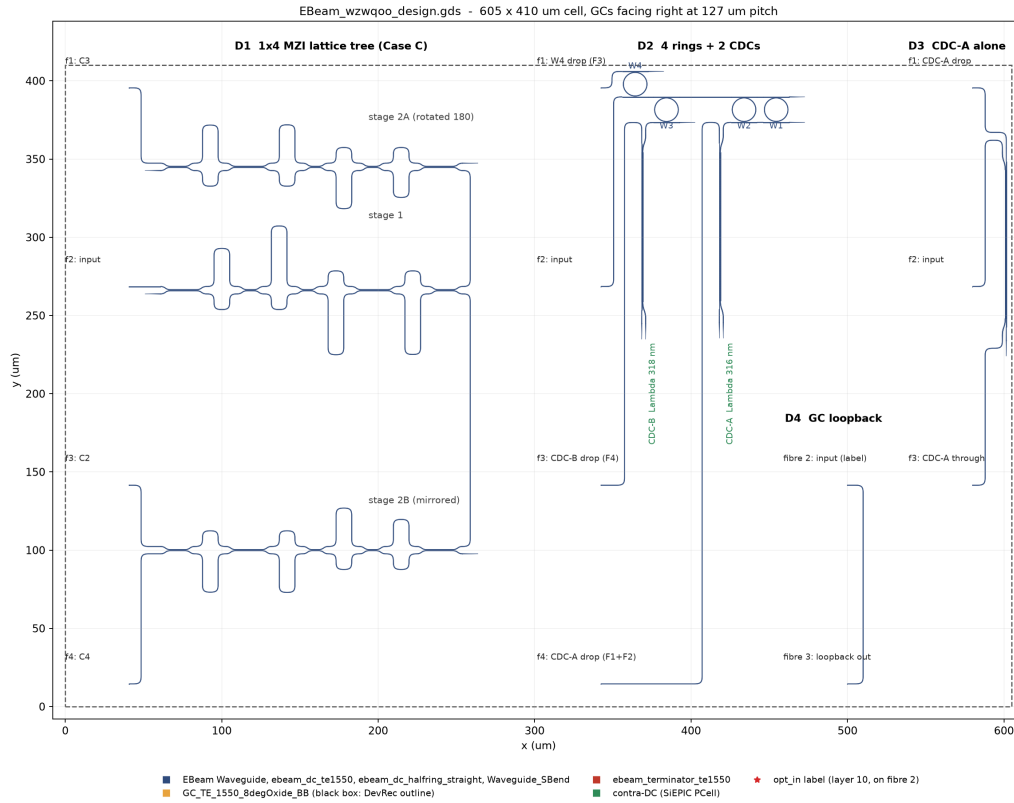


Fig. 5. Chip 1 (EBeam_wzwqoo_design). D1 lattice tree; D2 four rings and two CDCs; D3 CDC-A alone; D4 GC loopback.

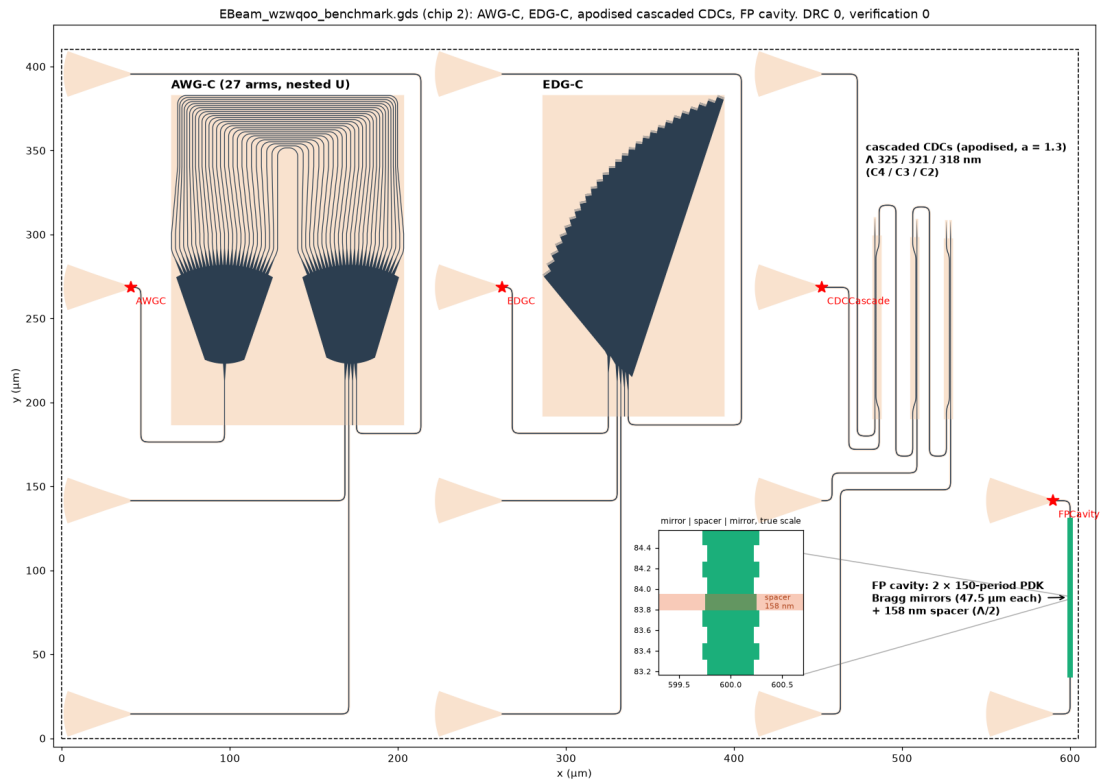


Fig. 6. Chip 2 (EBeam_wzwqoo_benchmark). AWG-C, EDG-C and three apodised cascaded CDCs, each with its own grating-coupler column, and the FP cavity at the right edge. Inset: the 158 nm spacer between the two Bragg mirrors at true scale.

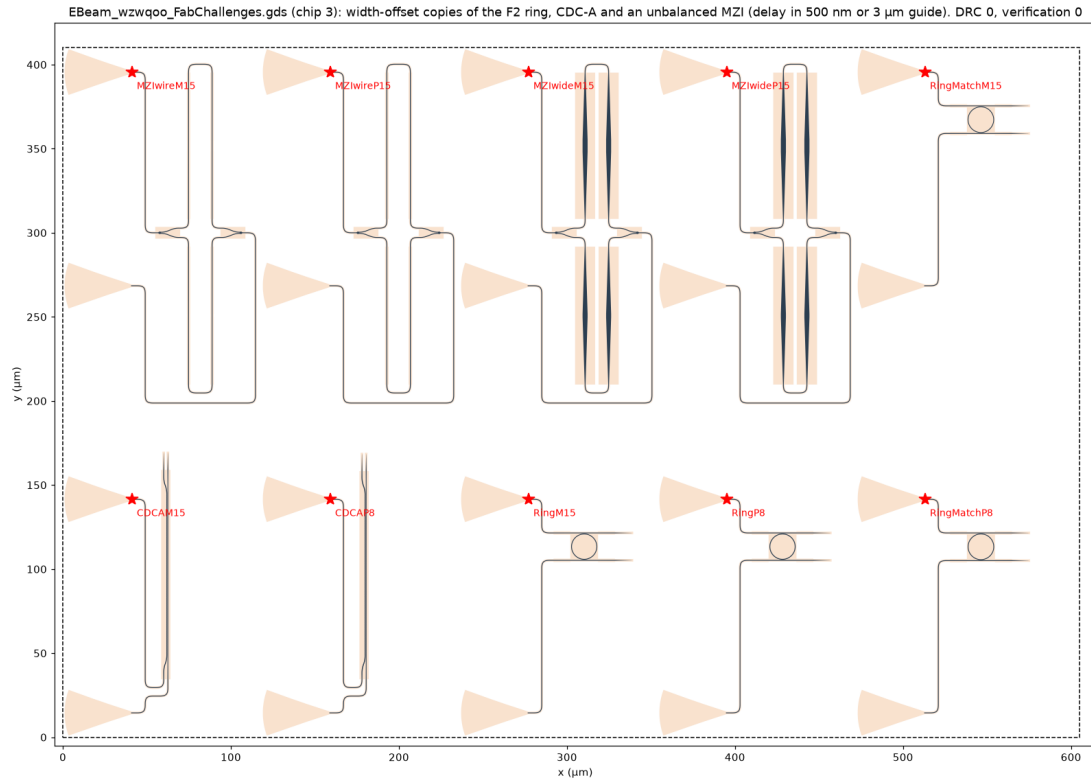


Fig. 7. Chip 3 (EBeam_wzwqoo_FabChallenges). Top: the two wire MZIs, the two 3 μm MZIs and the matched ring at -15 nm. Bottom: CDC-A at -15 and $+8$ nm, the 500 nm ring at -15 and $+8$ nm, and the matched ring at $+8$ nm.

carries; “—” is a fibre position with no coupler. Four devices measure only 3 of their 4 coarse channels, and the missing channel differs from device to device.

Table 13. Fibre-array map: fibre 1 (top) to fibre 4 (bottom) of each grating-coupler column

Chip	Device (label)	Fibre 1	Fibre 2	Fibre 3	Fibre 4	Not measured
1	LatticeD1	C3	input	C2	C4	C1 (terminated)
1	RingCDCD2	F3 (W4 drop)	input	F4 (CDC-B drop)	F1 + F2 (CDC-A drop)	—
1	CDCA	CDC-A drop	input	CDC-A through	—	—
1	GCLoopback	—	input	output	—	—
2	AWGC	C4	input	C1	C2	C3 (terminated)
2	EDGC	C1	input	C4	C3	C2 (terminated)
2	CDCCascade	C4	input	C3	C2	C1 (no fourth CDC)
2	FPCavity	—	—	input	transmission	—
3	MZIwire..., MZIwide..., RingMatchM15 (top pair)	input	MZI output / ring drop	—	—	—
3	CDCA..., Ring..., RingMatchP8 (bottom pair)	—	—	input	drop	—

The AWG-C and EDG-C channel order follows from the simulated output order and the layout: on the chip, AWG-C's outputs run C1, C2, C3, C4 from left to right and EDG-C's run C4, C3, C2, C1.

5.2. Verification

Simulations added after review.

- **MZI delay legs stay single-mode (chip 3).** Each 3 μm leg is reached through a 40 μm PDK taper ($0.5 \rightarrow 3 \mu\text{m}$), and a 3 μm guide carries seven TE modes. An EME of the PDK taper itself shows that TE0 keeps **99.8 %** of its power (-0.007 dB over 1500–1600 nm). Only **0.16 %** (-27.8 dB) is converted, almost all into TE2, and the odd modes are not excited, as the taper's symmetry requires.

- **The two legs of an arm do not couple (chip 3).** The two legs of an arm are 14 μm apart (an 11 μm gap). FDE supermodes of two 3 μm guides show their coupling falling about 160 times per 0.5 μm of gap: even at a 2 μm gap, a whole 87 μm leg would exchange only about 10^{-8} (−80 dB), and at 11 μm the coupling is far below anything measurable.
- **Cascade CDCs with the shallow teeth removed (chip 2).** The apodised cascade gratings keep every tooth at least 5 nm deep, an assumed print limit. As the worst case, the C2 grating (Λ 318 nm) was solved again by whole-grating EME with every tooth shallower than **10 nm** removed, as if those teeth did not print at all. The drop stays at **1546.4 nm** with **−0.21 dB** loss (−0.12 dB as drawn) and a 9.2 nm band (9.6 nm). The largest side lobe falls from **−13.2 to −16.2 dB**, and the worst leak within ± 3 nm of the neighbouring channels changes by **less than 0.5 dB**. Only the drop at the neighbouring channel centres rises, from −32 to −24 dB. The cascade's isolation therefore does not depend on the shallowest teeth printing.
- **Drawn widths of the width-offset copies (chip 3).** A width that is an odd number of nanometres is drawn **1 nm narrower** than specified, because each edge rounds inward on the 1 nm grid. The 500 nm ring's −15 nm copy is 484 nm instead of 485 nm, the matched ring's +8 nm copy 486 nm instead of 487 nm, and the MZI legs 484 / 514 nm and 2984 / 3014 nm instead of 485 / 515 and 2985 / 3015 nm; CDC-A's −15 nm copy has guides of 546 / 424 nm instead of 545 / 425 nm. Both MZI copies shift together, so their 30 nm span is unchanged. For the rings the span becomes **24 nm** (500 nm ring) and **22 nm** (W^*) instead of 23 nm, about **4 %** on the slope, so the analysis uses the drawn widths.

6. Design for Test

- **One input, up to three outputs.** Every device is a column of PDK TE grating couplers (GC_TE_1550_8degOxide_BB) at 127 μm pitch. The opt_in label is on the input coupler; the other couplers of the column are the outputs.
- **Labels.** All 18 labels follow opt_in_TE_1550_device_wzwqoo_<label> and are unique across the three files (Table 2).
- **Unused ports** end in PDK terminators (ebeam_terminator_te1550).
- **Missing outputs.** With only three output fibres, D1 routes out C2, C3 and C4 (not C1); AWG-C and EDG-C route three of their four outputs (the third from the left ends in a PDK terminator, so it leaves no open facet to reflect light back into the device); and the CDC cascade carries three CDCs (C4, C3, C2). The FP cavity uses two couplers, input on fibre 3 and output on fibre 4.
- **Calibration.** GCLoopback measures the grating-coupler pair; its spectrum is divided out of every other device. CDCA measures CDC-A alone, so the CDC band can be told apart from the ring lines in RingCDCD2. For chip 3, the F2 ring in RingCDCD2 and CDCA are the 0 nm references for the ring and CDC copies; the MZIs have no 0 nm copy and are read from the slope between their two copies.

Analysis plan.

1. Divide every spectrum by the GCLoopback spectrum.
2. LatticeD1, AWGC, EDGC, CDCCascade: read each output's peak, 1 dB band and worst leak within ± 3 nm, and compare them with Table 12.
3. RingCDCD2: fit each ring line (centre, FWHM), read n_g from the ring FSR by (5), and the isolation between outputs.
4. FPCavity: line position and width.
5. Chip 3: Section 7.

7. Fabrication and Temperature Experiments

Experiment 1: does the F2 ring stay inside CDC-A's band? On chip 1 the F2 line sits only 0.48 nm inside CDC-A's upper 1 dB edge. By (1), a width error d moves both, but at different rates: 0.591 nm per nm of width for the ring and 0.667 nm/nm for the CDC. The margin is used up at $d \approx -6$ nm. A ring 479 nm wide (W^*) moves at 0.662 nm/nm, nearly as fast as the CDC, and keeps F2 at every offset.

Experiment 2: a delay line that ignores width. In an MZI everything the two arms share (bends, splitters, equal straights) cancels. Only the extra length ΔL matters, so only the width of the guide that carries ΔL moves the fringes. Putting ΔL in a 3 μm guide instead of 500 nm wire makes the fringes 167 times less sensitive in simulation. $\Delta L = 10 \mu\text{m}$ gives an FSR of about 57 nm, so each spectrum shows one or two nulls. The experiment tracks the position of one null, which moves by less than half an FSR over the ± 15 nm range, so the same null is identified on every copy.

How chip 3 is read. The two copies of each device share the same unknown fabrication bias, so the slope between them (wavelength shift divided by the 23 or 30 nm width difference) does not depend on that bias.

Table 14. Chip-3 predictions (INTERCONNECT)

Device	$d = -15$ nm	$d = 0$	$d = +8$ or $+15$ nm
MZI null, delay in 500 nm wire	1527.06 nm	1536.03 nm	1544.20 nm (+15)
MZI null, delay in 3 μ m guide	1568.09 nm	1568.14 nm	1568.19 nm (+15)
CDC-A 1 dB band	1526.23–1534.83 nm	1536.40–1545.00 nm	1541.58–1550.18 nm (+8)
F2 ring line, 500 nm	1535.44 nm (outside the band: F2 lost)	1544.52 nm	1549.02 nm (+8)
F2 ring line, 479 nm (W^*)	1534.37 nm (0.46 nm inside)	1544.53 nm	1549.59 nm (+8)

The $d = 0$ column is simulated; the chip-1 devices provide it on the chip for the 500 nm ring and CDC-A only.

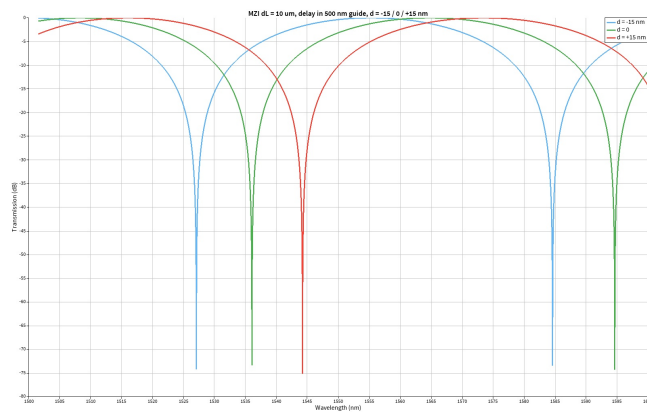


Fig. 8. MZI with its delay in 500 nm wire, $d = -15 / 0 / +15$ nm: the null moves about 9 nm per 15 nm of width.

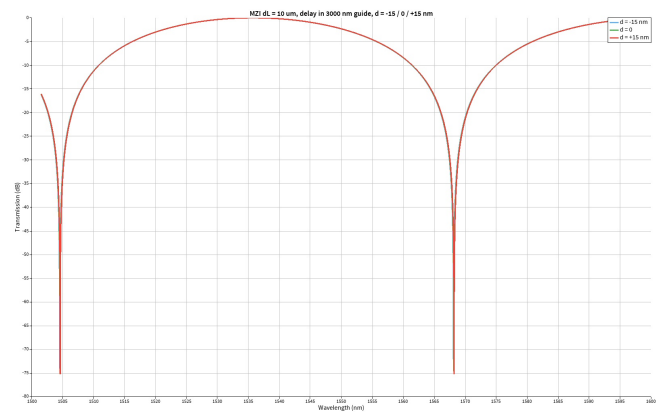


Fig. 9. The same MZI with its delay in a 3 μ m guide: the three curves lie within 0.10 nm.

Limits of the width-offset reading.

- **Two copies give one slope, which assumes a linear response.** The simulated responses bend slightly. Table 15 compares the slope on each side of $d = 0$ with the two-point slope the experiment measures: they differ by up to ± 5 %.
- **Shared bias.** Both copies sit on the same die, so an unknown fabrication bias b shifts both widths alike. It cancels in the difference, but the two-point slope is then the slope at b instead of at 0. With the curvature of Table 15, a bias of **10 nm** changes the measured slope by **4–6 %**, and the ring-to-CDC slope difference (0.077 nm/nm) by ± 0.011 nm/nm. The wire-to-wide ratio of the MZIs (about $170\times$) is far larger than these errors.
- **No $d = 0$ copy of the MZIs.** Neither MZI has a nominal-width copy on any chip, so their slope rests on two points and the simulated null (1536.03 nm) is the only $d = 0$ reference. The curvature cannot be checked on the chip for the MZIs, and the bias cannot be read from them. The rings and CDC-A have their $d = 0$ references on chip 1, so they give three points.

Table 15. Slope of each chip-3 feature on either side of $d = 0$ (from Table 14), and how much a shared 10 nm bias moves the two-point slope

Feature	Slope $-15 \rightarrow 0$ nm	Slope $0 \rightarrow +8 / +15$ nm	Two-point slope (measured)	Change per 10 nm of bias
MZI null, delay in 500 nm wire	0.598 nm/nm	0.545 nm/nm (+15)	0.571 nm/nm	0.036 nm/nm (6 %)
MZI null, delay in 3 μ m guide	0.0033 nm/nm	0.0033 nm/nm (+15)	0.0033 nm/nm	< 0.001 nm/nm
F2 ring line, 500 nm	0.605 nm/nm	0.563 nm/nm (+8)	0.590 nm/nm	0.037 nm/nm (6 %)
F2 ring line, 479 nm (W^*)	0.677 nm/nm	0.633 nm/nm (+8)	0.662 nm/nm	0.039 nm/nm (6 %)
CDC-A upper 1 dB edge	0.678 nm/nm	0.648 nm/nm (+8)	0.667 nm/nm	0.027 nm/nm (4 %)

Experiment 3: temperature. openEBL can measure devices a second time, about 10 K warmer. RingCDCD2, CDCA and FPCavity are requested. By (1), every feature should move by about +0.70 nm, and the ring and CDC by the same amount to within 0.005 nm. The measured shift gives the thermo-optic coefficient of the fabricated guide:

$$\frac{dn_{\text{eff}}}{dT} = \frac{n_g}{\lambda} \frac{\Delta\lambda}{\Delta T} \quad (14)$$

8. Test Results

Measurements pending: this section will hold the openEBL results.

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